Question 1:

Which of the following decision(s) is(are) related to managing a company's capital structure?

I. Company ABC choosing to refinance its existing debt with a lower interest rate bond II. Company ABC opting to retain earnings rather than paying dividends III. Company ABC selecting to lease office space instead of purchasing it IV. Company ABC deciding to increase the issuance of preferred stocks
Type: multiple-choice

Options:

a. I only

b. II only

c. III only

d. I and II only

e. I and IV only

f. II and III only

Question 2:

Which of the following statements (s) is (are) accurate regarding different business structures? I. In an LLC, members have limited liability for the company's debts II. A disadvantage of a sole proprietorship is that the owner has unlimited personal liability III. Corporations face more difficulty in transferring ownership compared to partnerships IV. Limited partnerships require at least one general partner who manages the business and assumes full liability V. Raising capital is easier for corporations due to their ability to issue stocks VI. Partnerships offer tax benefits as profits pass through directly to partners' personal income

Type: multiple-choice

Options:

a. II, III, IV and VI only

b. I, II, IV, and V only

c. II, III, IV, and VI only

d. I, II, IV, and VI only

e. I, III, V, and VI only

f. All statements are true

Question 3:

Which of the following statement(s) is(are) true about a premium bond? I. The price of the bond will increase over time as it approaches maturity, given all else unchanged II. Before the bond matures, the price of the bond is higher than its face value III. The coupon rate of the bond is higher than the yield to maturity IV. Bond value will decrease if market interest rates rise, given all else unchanged V. Value of the bond would increase if the yield to maturity decreases, given all else unchanged

Question 4:

Which of the following bonds has the least interest rate risk? Bond X matures in 8 years with an annual coupon payment of $60 and a face value of $1,500. Bond Y is a zero-coupon bond maturing in 5 years. Bond Z offers an annual coupon of $45 for 7 years before maturity, with a face value of $2,000. Bond W will mature in 12 years and has a coupon rate of 8%. Bond V matures in 9 years and pays out an annual coupon of $30 on a face value of $1,000.

Type: multiple-choice

Options:

a. Bond X

b. Bond Y

c. Bond Z

d. Bond W

e. Bond V

f. Not enough information is available

Question 5:

Question: A financial institution aims to charge an EAR of 12% on their loans. If monthly compounding is applied, what APR should the institution report? Choose the closest answer.

Type: multiple-choice

Options:

a. 11.5%

b. 11.7%

c. 11.9%

d. 12.0%

e. 12.4%

Question 6:

Stock XYZ is also a constant growth stock with an indefinite dividend payout plan. A dividend of $3 was just paid today, and the current price of Stock XYZ is $280. One year from now (after Year-1’s dividend payment), the stock price will be $294. What are the required return, capital gains yield, and dividend yield for Stock XYZ? Choose the closest answer.

Type: multiple-choice

Options:

a. 7%, 5%, 1%

b. 6%, 3%, 2%

c. 8%, 4%, 2%

d. 5%, 2%, 3%

e. 7.5%, 4%, 1%

Question 7:

Given the target question, here's an analogous one:

A bond with a BBB credit rating has a price of $1809.25 and matures in 15 years. It pays semi-annual coupons. If the yield to maturity (YTM) for this bond is 7%, what is the annual coupon payment? Choose the closest answer.
Type: multiple-choice

Options:

a. $45.00

b. $72.50

c. $98.00

d. $123.50

e. $150.00

Question 8:

Given the details about Stock Alpha and Stock Gamma, create a similar scenario for two other stocks, Beta and Delta. Stock Beta just paid out a dividend of $40 today with an expected growth rate of 3% annually over the next 20 years, followed by constant dividends thereafter. The required return is 8%. Stock Delta will pay a quarterly dividend of $10 starting today and continue indefinitely. If the current market price for Stock Beta is $675, which of the following statements are true regarding these two stocks?

I. The annual percentage rate (APR) on Stock Beta is lower than that of Stock Delta.

II. The present value of Stock Beta is greater than the current market price.

Type: multiple-choice

Options:

a. None of the statements are true

b. I only

c. II only

d. Both I and II are true

Question 9:

You just invested $360,000 at the start of the year in an annuity that will pay you $45,000 at the end of each year for 20 years, with the first payment received one year from now. If it takes exactly 20 years to recover your initial investment through these annual payments, what is the approximate APR of this annuity? Choose the closest answer.

Question 10:

Given the target question 'Question: Which of the below statement(s) is(are) true regarding the term structure of interest rates? I. If the yield curve is currently upward sloping, an increase in long-term real interest rates relative to short-term real interest rates will steepen the slope further. II. Assuming a downward-sloping yield curve with constant real interest rates across maturities, this scenario suggests that long-term nominal interest rates are lower than what would be expected if inflation expectations were uniform over time. III. In a situation where the term structure is downward sloping and the real rate is stable, it implies

that investors demand a higher premium for short-term investments compared to long-term ones.
Type: multiple-choice

Options:

a. I and II only

b. II and III only

c. I, II and III

d. None of the statements are true

Question 11:

Assume today is July 1, 2023. You are planning for your retirement on January 1, 2045, and you have two retirement goals. Goal 1: Three months after you retire, you want to purchase a luxury car costing $100,000 at the beginning of the third month that you retire. Goal 2: Once you retire, at the end of each month you are going to withdraw some money for daily and medical expenses. Your first withdrawal will be at the end of the month that you retire. For your first withdrawal, you want to take $15,000, and after that, you expect that the expenses are going to grow by 0.3% each month and last for 28 years. APR is 7% compounded monthly. a. To fulfill your two retirement goals, what is the total amount that you need to have saved by the time you retire? b. On average your salary is expected to grow at a rate of 0.15% each month, and you are going to deposit 25% of your monthly salary into a savings account. If by the time you retire you have saved $Y in total (amount as calculated in part (a)), what is your monthly salary when you make your first deposit? What is your monthly salary when you make your third deposit?

Question 12:

Company XYZ currently has Bond X in the market. Bond X is a 10% 20-year bond which pays out semi-annually. Bond X has a face value of $5000. Company XYZ just issued a new Bond - Bond Y. Bond Y sells as a discount bond, at 8% below par. Bond Y pays out coupon every year and matures in 10 years. The YTM for both bonds today is 6%. a. What is the current yield of Bond X? b. What is the coupon rate of Bond Y? c. 5 years from now, YTM decreases to 3% for both bonds, and you sell Bond X, what is the holding period return for Bond X?

Question 13:

A manufacturing firm, XYZ Inc., plans to start a major project next month with a partner company, YZL Corp. The total cost of the project will be $20 million today, and it is scheduled for completion exactly five years from now. Upon successful completion, YZL Corp has agreed to pay XYZ Inc. $25 million in a single payment two months after the project ends. Seventy percent of the project costs are covered by a 30-year loan at an annual percentage rate (APR) of 4%. The loan requires monthly fixed payments starting one month from now.

a. Calculate the monthly repayment amount for XYZ Inc.

b. Immediately following receipt of payment from YZL Corp and making the last regular monthly installment, XYZ Inc. intends to settle all remaining loan obligations in a lump sum. Determine the size of this final balloon payment.

Question 14:

Given the details about dividends paid by stock ABC issued by Company ABC, where each dividend is paid on January 1st, summarize the payment pattern from 2002 to 2012 and calculate the price of the stock ABC on January 1, 2002. In 2002, a dividend of $5 was paid, followed by a reduction of $1 per year for three years due to financial preparation for an expansion project. No dividends were paid during the project period from 2006 to 2008. Upon completion in 2009, a $3 dividend was issued with a growth rate of 4% increasing by 1 percentage point annually until it reaches 7%, after which it grows at 7% indefinitely. Assume the required return for stock ABC is 8%.